

Gerson Vidal Alcántara

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Software engineer (expected graduation: January 2027), specialized in backend development with Java and Spring Boot. I have experience developing REST APIs for projects in different fields, including healthcare, project management, and distributed systems for the Internet of Things. Additionally, I have experience with data persistence using Hibernate/JPA, SQL databases, Docker, implementation of SOLID principles, software design, testing, and backend system deployment.

RELEVANT EXPERIENCE

Vida Rosa

Software Engineer | Backend Developer

Ixtaczoquitlán, Mexico

August 2025 – June 2026

- Redesigned and refactored the entire backend architecture and implementation to adapt it as a data collection tool, carrying out a software engineering process that included planning, analysis, design, development, testing, deployment, and maintenance.
- Ensured reliable data collection, with more than 80 records processed by the REST API and successfully stored in a PostgreSQL database, through backend preparation and validation, anticipating incorrect data-entry scenarios.
- Achieved a 100% test success rate and an average code coverage of 92.1% across the microservices, by designing and executing more than 250 unit tests and integration tests, with results analyzed using Allure Report and SonarQube.
- Strengthened backend reliability and reduced coupling by designing and implementing a validation architecture in a new layer, focused on facilitating validation extension and reuse.
- Improved code quality, achieving an analysis with no pending findings in SonarQube by resolving the detected issues.
- Deployed the system to production, fully configuring a VPS with HTTPS, a firewall, NGINX, and backend deployment with Docker and Docker Hub.

IoT Greenhouse System

Software Engineer | Software Developer

Ixtaczoquitlán, Mexico

March 2025 – June 2025

- Avoided code duplication and simplified the extension of sensors and actuators by designing a class hierarchy that reuses common attributes and behaviors, as well as by applying the Factory Method pattern.
- Standardized a communication strategy to enable dynamic control and accurate device feedback, by defining specific MQTT topics for commands, data, mode and status, together with its implementation in code.

Critical Path Method Calculation Application

Software Engineer | Backend Developer

Ixtaczoquitlán, Mexico

October 2024 – December 2024

- Prevented implementation errors by designing the technical foundation of the system before coding, including the software architecture, use cases, backend REST API, and data model.
- Implemented the complete backend solution using Java, Spring Boot, and Hibernate for the REST API and PostgreSQL for the database.
- Avoided unnecessary storage in the database, by persisting the textual sequence of the critical path and generating its diagram on demand from the backend.

EDUCATION

UNIVERSIDAD VERACRUZANA

Bachelor's Degree in Software Engineering

Ixtaczoquitlán, Mexico

August 2022 – January 2027

(expected)

SKILLS

- Java, Spring Boot, REST APIs, Hibernate/JPA, Python, Flask, SQL, PostgreSQL, SonarQube, Docker, Docker Hub, Git, GitHub, Object-Oriented Programming, SOLID Principles, Design Patterns, Unit Testing, Integration Testing, GNU/Linux, VPS Deployment, Data Structures, and Algorithms.
- Problem Solving, Analytical Thinking, Collaboration, Organization, Continuous Improvement.
- Spanish (native), English (advanced).